

Deployment & System Architecture Guide

AWS EC2, TERRAFORM, DOCKER COMPOSE & NGINX SETUP

1. System Specifications & Production Environment

- **Host Platform:** AWS EC2 t3.small (2 vCPU, 2 GiB RAM, x86_64, Ubuntu 24.04 LTS)
- **Root Storage:** 40 GiB gp3 EBS encrypted root volume
- **Memory Safety Buffer:** 2 GB /swapfile configured on EBS
- **Public Gateway:** Nginx reverse proxy on Port 80 (Internal proxy to 127.0.0.1:7860)
- **Process Management:** Docker Compose with restart: unless-stopped

2. Method 1: Infrastructure as Code (Terraform)

```
cd terraform cp terraform.tfvars.example terraform.tfvars # Set admin_cidr_blocks = ["YOUR_PUBLIC_IP/32"] terraform init
terraform apply
```

3. Method 2: Docker Compose (Local or Host)

```
git clone https://github.com/Aadithya2222/Logistics-Object-Detection-Model.git cd Logistics-Object-Detection-Model docker
compose up -d --build
```

4. Security Group Rules

- **TCP 22:** SSH restricted to administrator IP only.
- **TCP 80:** HTTP open for public web API access.
- **TCP 443:** HTTPS open (Reserved for future SSL configuration).
- **TCP 7860:** Isolated internally (NOT exposed publicly).